

# AI Career Guidance & Learning Assistant

Minimum Viable Product (MVP) Comprehensive Architecture & Feature Specifications

## PRODUCT REQUIREMENTS DOCUMENT

### SYSTEM EXECUTION SEQUENCE

Interest  
Analysis



Career  
Recommendation



Roadmap  
Generation



Course  
Matcher



PDF  
Assistant



AI  
Tutor

## 1. AI Career Discovery

The foundation of the platform relies on mapping user profiles to modern technical and non-technical fields. The system collects granular metrics from the user and passes them through an analytical model to extract recommendations.

- **Dynamic Assessment:** User inputs data regarding personal interests, long-term professional goals, favorite academic subjects, and current baseline skills.
- **Matching Engine:** The AI evaluates compatibility parameters against industry roles, explaining exactly *why* each career path aligns with the user's specific profile.

### Primary Recommended Career Paths:

AI Engineer

Data Scientist

Full Stack Developer

Cybersecurity Analyst

UI/UX Designer

Cloud Architect

DevOps Specialist

## 2. Personalized Career Roadmap

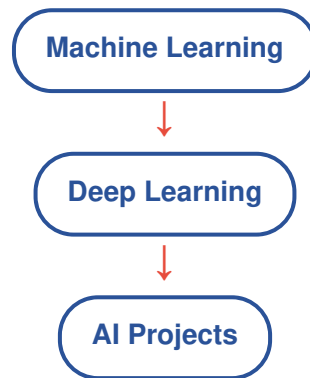
Once a target career path is established, the platform synthesizes an structured learning path tailored to the user's current baseline. The curriculum maps sequential progression across standardized milestones (Beginner, Intermediate, and Advanced stages) and dynamic adapts as the user submits progress data.

Python



Data Structures





### 3. Course Recommendation Engine

To fulfill the objectives established in the user's roadmap, the system indexes external educational platforms to recommend precise learning modules based on explicit constraints: **Career Goal**, **Skill Level**, and **Financial Budget**.

#### Free Resources

Curation focuses on high-quality open-source curriculums, standard documentation tracks, and community-driven video guides.

#### Paid Resources

Prioritizes structured professional certificates, bootcamps, and specialized courses offering verified credentials.

#### Supported Core Platforms:

[YouTube](#)[Coursera](#)[Udemy](#)[freeCodeCamp](#)[Kaggle](#)

### 4. AI PDF & Notes Assistant

An on-demand ingestion system enabling contextual file learning. Users upload reference documents including PDFs, lecture slides (PPT), or raw Markdown text notes to extract structured secondary learning material through LLM analysis.

- **Automated Summarization:** Distills comprehensive multi-page documents into high-yield, structured takeaways.
- **Question Generation:** Produces conceptual question sets, custom Multiple Choice Questions (MCQs), and interactive flashcards to optimize active recall.
- **Contextual Deep Dives:** Acts as an on-demand explainer for complex, ambiguous definitions or mathematical formulas found within the text.

## 5. AI Tutor

A round-the-clock interactive chat interface optimizing personal study. It delivers immediate code explanations in clear, accessible prose.

- Instant multi-lingual concept translations
- Code debugging and refactoring walkthroughs
- Personalized pacing based on user questions

## 6. Project Recommendation

Generates actionable, practical development project suggestions mapped directly to the user's current roadmap stage to build a robust professional portfolio.

- **Beginner:** Titanic Data Prediction
- **Intermediate:** Movie Recommendation Engine
- **Advanced:** AI Chatbot, Resume Screening AI

## Future Roadmap (Version 2.0 Features)

Planned updates to transform the MVP from a standalone learning guide into an end-to-end career transition tool:

- **Automated Resume Analyzer:** Real-time applicant tracking system (ATS) optimization.
- **Interview Preparation Suite:** Automated mock interviews with detailed feedback loops.
- **Live Job & Internship Matching:** Real-time opportunity pipeline ingestion.
- **AI Voice Mentor:** Interactive, speech-driven learning and conversational training.